



File View Settings Help



Python 3.13 (XPython)

```
fan_low_level = low_level
fan_medium_level = medium_level
fan_high_level = high_level

# Step 5: Aggregate fuzzy outputs
activated_low = np.minimum(fan_low_level, fan_low)
activated_medium = np.minimum(fan_medium_level, fan_medium)
activated_high = np.minimum(fan_high_level, fan_high)

aggregated = np.maximum(
    activated_low,
    np.maximum(activated_medium, activated_high)
)

# Step 6: Defuzzification using centroid method
if np.sum(aggregated) == 0:
    fan_output = 0
else:
    fan_output = np.sum(fan_speed * aggregated) / np.sum(aggregated)

# Step 7: Display output
print(f"Temperature: {temperature_input}°C")
print(f"Fuzzified fan speed: {fan_output:.2f}")

Temperature: 28°C
Fuzzified fan speed: 7.17
```

[]: